

EMMANUEL BOLARINWA

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EDUCATION

CARNEGIE MELLON UNIVERSITY AFRICA

2026 – 2028

M.S. Engineering Artificial Intelligence

Kigali, Rwanda

Expected graduation 2028.

OBAFEMI AWOLOWO UNIVERSITY

2017 – 2023

B.Sc. Computer Engineering

Ile-Ife, Nigeria

Second Class Honours (Upper Division). Ranked 3rd nationally.

Thesis – Design and implementation of a low-cost electronic nose system: a sensor device that detects and classifies carbon monoxide and ethylene, using support vector machines, gradient boosting and principal component analysis. Published as a peer-reviewed paper.

Coursework – Artificial Intelligence, Natural Language Processing, Data Structures and Algorithms, Database Management Systems, Numerical Computation, Expert Systems, Software Engineering, Computer Networks.

RESEARCH EXPERIENCE

LYNGUAL LABS

Nov 2024 – Present

Artificial Intelligence Researcher

Lagos, Nigeria

- Co-led the collection and transcription of 100 hours of Yoruba-English code-switched speech across 10+ domains, creating one of the first open corpora for this language pair.
- Fine-tuned and benchmarked multilingual speech recognition models on code-switched speech, beating monolingual baselines and producing three peer-reviewed papers in 2025.
- Handled preprocessing, segmentation and alignment of the multilingual audio, and designed the team's evaluation strategy.
- Exploring how these models can drive voice interfaces for open-source energy planning tools.

PERVASIVE AND MOBILE COMPUTING RESEARCH GROUP

2023 – 2024

Undergraduate Researcher, supervised by Dr. S. Aina

Ile-Ife, Nigeria

- Reviewed the state of the art to guide the design of a low-cost electronic nose, then built the machine learning pipeline, from sensor data through principal component analysis to gas classification with gradient boosting.

MICROELECTRONICS RESEARCH GROUP

Jul 2021 – Dec 2021

Research Intern

Ile-Ife, Nigeria

- Reviewed machine learning research on gas sensors and tuned several algorithms for a gas sensor dataset, improving accuracy and inference speed.
- Deployed a real-time facemask detection model to the SIPEED Maixduino edge device.

PROFESSIONAL EXPERIENCE

OPEN ENERGY TRANSITION

2023 – Present

Energy Systems Modeler

Bayreuth, Germany (Remote)

- Built concentrated solar power modeling with thermal storage and a Monte Carlo simulation module into PyPSA-Earth, letting users quantify uncertainty in national energy plans.
- Ran a comparative geographic analysis of ENTSO-E and OpenStreetMap grid data to move the framework onto fully open data.
- Contributed to a fully sector-coupled model of the United States energy system.

HAMOYE

Jun 2022 – Oct 2022

Data Engineering and Machine Learning Operations Intern

Lagos, Nigeria

- Built an end-to-end machine learning project for quantitative data analysis that placed in the top five of the cohort on design and implementation.

ZURI

2021

Backend Engineering Intern

Lagos, Nigeria

- Built and maintained a Django blog application, covering server-side logic, user authentication and database management.

XLEVELRETAIL

2020 – 2023

Web Developer

Lagos, Nigeria

- Led development of four e-commerce sites from scratch in WordPress, integrating WooCommerce and custom features.

TEACHING EXPERIENCE

LYNGUAL LABS ACADEMY

2025 – Present

Machine Learning Tutor

Remote

- Teach live sessions and hands-on labs on Python, data science foundations, feature engineering and model interpretation.
- Mentor students through the full machine learning lifecycle, from data wrangling to model tuning and experiment tracking with MLflow, ending in portfolio-ready projects.

INTERNATIONAL CENTRE FOR HYDROPOWER WORKSHOP

2024

Invited Lecturer

Nairobi, Kenya

- Designed and delivered a lecture on modeling energy system uncertainties with Monte Carlo simulation to professionals and academics from eight African countries, as the youngest invited lecturer in the organisation's history.
- Ran the hands-on sessions, guiding participants to build transparent, replicable energy models with PyPSA, Snakemake and geographic information systems.

IJAN AFRICA

2023 – 2024

Data Analysis Instructor

Lagos, Nigeria

- Designed and taught a four-month project-based curriculum in SQL, Python and Excel for 20+ students, mostly working professionals.
- Coached students to build portfolios and move into paid data roles.

OBAFEMI AWOLOWO UNIVERSITY

2023 – 2024

Teaching Assistant

Ile-Ife, Nigeria

- Taught Python programming to over 100 students through a mix of virtual lectures and in-person lab guidance.

PUBLICATIONS AND TALKS

2025 Bolarinwa, E., Babatunde, O., Olufemi, V., Moshood, K., & Williams, O. *Closing the Gap in Low-Resource ASR: Leveraging Multilingual Models for Code-Switched Yoruba-English Speech*. Deep Learning Indaba 2025; to appear in PMLR.

2025 Babatunde, O. B., Olufemi, V. T., Bolarinwa, E. T., Moshood, K. Y., & Emezue, C. C. *Beyond Monolingual Limits: Fine-Tuning Monolingual ASR for Yoruba-English Code-Switching*. 7th Workshop on Computational Approaches to Linguistic Code-Switching, co-located with NAACL 2025.

2025 Babatunde, O. B., Olufemi, V. T., Bolarinwa, E. T., & Moshood, K. Y. *Challenging Multimodal LLMs with African Standardized Exams: A Document VQA Evaluation*. AfricaNLP Workshop, co-located with ACL 2025.

2025 Co-presented a technical talk on using PyPSA-Earth and OpenStreetMap to build transparent, replicable models for sustainable energy planning in Africa. Open Source Conference Africa.

2024 Aina, S., Gambo, I. P., Ogungbe, O. A., Bolarinwa, E., & Oluwaranti, A. I. *A Low-Cost Electronic Nose Device*. International Journal of Information Research and Review, 11(5), 7774–7777.

PROJECTS

AFRILEARNER360 | *Python, large language models, React, TypeScript*

2026

- Co-built a low-resource system that profiles primary school learners by how they engage with content and adapts curriculum material to local culture, using a public primary school in Rwanda as the case study.
- Led the learner profiling work and its integration with content delivery, pairing a language model for question generation with deterministic scoring so repeated assessments stay consistent.

YORUBA-ENGLISH CODE-SWITCHING SPEECH CORPUS | *Python, audio processing*

2025

- Curated a code-switched speech dataset through systematic collection, segmentation, language-boundary marking and bilingual transcription, giving Yoruba speech research usable training data.

PYPASA HELPER BOT | *Python, Discord API, retrieval*

2025

- Built a Discord bot that answers setup and modeling questions for the global PyPSA community, cutting the time maintainers spend on repeat questions.

SOLSIGHT SERVERLESS PIPELINE | *Python, Amazon Web Services, GitHub Actions*

2024

- Architected the ingestion, cleaning and transformation logic for an enterprise serverless data pipeline, and integrated it into a continuous deployment workflow.

GITHUB ARCHIVE DATA PIPELINE | *Terraform, Prefect, PySpark, dbt, BigQuery*

2023

- Engineered an end-to-end batch pipeline on Google Cloud, from infrastructure provisioning through transformation, delivering developer activity analytics in Looker Studio dashboards.
- PODCAST SUMMARIZER** | *Python, Streamlit, Modal Labs* 2023
- Fine-tuned a summarization model for podcast transcripts and shipped it behind an interactive front end with summaries, guest details, highlights and chapters.

AWARDS AND RECOGNITION

- 2025** Paper Award, Deep Learning Indaba, given to the top one percent of submissions at Africa’s largest machine learning gathering.
- 2024** Youngest invited lecturer in the history of the International Centre for Hydropower.
- 2024** Acknowledged in Xiong et al. (2024) for early development and validation work on integrating OpenStreetMap data into PyPSA-Earth.

SKILLS

- Programming** Python (advanced), SQL, Git, Bash
- Machine Learning** PyTorch, TensorFlow, Keras, scikit-learn, natural language processing, speech recognition, large language models, MLflow
- Energy and Geospatial** PyPSA, PyPSA-Earth, OSeMOSYS, Snakemake, GeoPandas, geographic information systems, Monte Carlo uncertainty modeling
- Data Engineering** PySpark, dbt, Prefect, Terraform, Docker, BigQuery, PostgreSQL, MySQL, LocalStack
- Cloud and Visualization** Google Cloud, Amazon Web Services, Azure, Grafana, Looker, Power BI, Matplotlib, Seaborn
- Languages** English, Yoruba

SERVICE AND VOLUNTEERING

- PYPSA-EARTH COMMUNITY** 2023 – Present
Maintainer and Documentation Lead Remote
- Review community contributions to keep the framework stable, and lead the overhaul of the project documentation to make it easier for new users to start.
- DATA EPIC** 2024 – Present
Data Engineering Mentor Remote
- Guide aspiring data engineers on tooling and best practice through hands-on projects.
- COWRYWISE** 2022 – 2023
Student Ambassador Lagos, Nigeria
- Promoted financial literacy among students and relayed community feedback into youth-focused product decisions.
- HYDEASHUB** 2016 – 2024
Chess Instructor Lagos, Nigeria
- Built chess programmes and ran tournaments for children in the local community.

CERTIFICATIONS AND MEMBERSHIPS

- Certifications** Data Engineering Zoomcamp (DataTalksClub, 2023); Building AI Products with OpenAI (Uplimit, 2023); Machine Learning with Amazon Web Services (Udacity, 2022); Neural Networks and Deep Learning (Coursera, 2022)
- Memberships** Graduate Member, Nigerian Society of Engineers; Member, Data Science Nigeria